

ARJUN SEDANI

Senior Full Stack Developer | React | TypeScript | Node.js | Python | Design Systems | Micro Frontends
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PROFESSIONAL SUMMARY

Full Stack Developer with 10+ years of experience building large-scale enterprise applications, reusable component libraries, and micro-frontend architectures. Proven track record delivering complex approval workflow platforms, energy trading dashboards, design systems, and e-commerce solutions for Fortune 500 clients (Shell, Nayara Energy, DBS Bank, Walmart, Nike). Expert in React/TypeScript ecosystem with hands-on experience architecting and publishing NPM component libraries (53+ React components, 78+ Preact components). Proficient in AI & Prompt Engineering with advanced experience across LLM models (GPT-5.3, Codex, Claude Opus, Claude Sonnet) for code generation, workflow automation, and intelligent tooling. Strong background in Node.js, Python, MongoDB, Redux, Module Federation, RBAC, and on-premises deployments. Scrum Certified.

TECHNICAL SKILLS

Frontend: React 18/19, TypeScript, JavaScript (ES2024), Next.js, Preact 10, Redux Toolkit, Redux Saga, React Router, React Native, Ant Design, styled-components, SCSS/Sass, Tailwind CSS, AG Grid Enterprise, Highcharts, PowerBI, CKEditor, React Quill, A-Frame (VR/AR), Web Components

Component Libraries: Storybook 10, Turborepo, pnpm Workspaces, Changesets, Figma Token Sync, Atomic Design, NPM Publishing, esbuild custom tooling

Backend: Node.js, Express.js, Python, Flask, REST APIs, GraphQL, WebSocket, Serverless Functions (Vercel/AWS Lambda)

Database: MongoDB Atlas, CosmosDB, Firebase, SQL Server, Vercel KV, NoSQL

Architecture: Micro Frontends (Module Federation), Design System Monorepo, Atomic Design, Component Registry Pattern, Serverless, PWA, RBAC

Testing: Jest, Vitest, React Testing Library, Cypress (Component + E2E), Playwright, Pact (Contract Testing), SonarQube

DevOps & Cloud: Azure (App Insights, Functions), AWS (Lambda, Lex), Vercel, GitHub Actions, Jenkins, Docker, CI/CD, On-Premises Deployment

Tools: Git, Bitbucket, JIRA, Storybook, Adobe Analytics, Figma, Webpack, Vite, esbuild

Auth & Security: SSO, OIDC, OAuth2, JWT, RSA Encryption, RBAC, OTP Authentication

AI & Prompt Engineering: GPT-5.3, OpenAI Codex, Claude Opus, Claude Sonnet, GitHub Copilot, AI-assisted code generation, LLM prompt design, Agentic workflows, Natural Language Processing

PROFESSIONAL EXPERIENCE

KPMG IndiaMay 2023 – Present

Assistant Manager – Technology Consulting

Bengaluru, Karnataka

Project: NFA Portal – Unified Approval System (Client: Nayara Energy)

- Architected and built a full-stack multi-module NFA (Note for Approval) portal handling 7 approval workflows — Procurement (6 sub-types), Marketing (6 sub-types), CRF/ARF, Tender, Legal, Credit Approval, and Payment — each with specialized forms, dashboards, and reporting.
- Developed role-based access control (RBAC) system with 15+ role IDs controlling route-level access, sidebar menu filtering, form editability, and module visibility per user.

- Built backend API integration layer with Axios interceptors, centralized error handling, and environment-aware API routing for .NET backend services.
- Created responsive layout with mobile breadcrumb navigation and Drawer-based sidebar; implemented approval workflow visualization with collapsible step panels and rich text editor integration.
- Engineered domain-specific Redux slices (nfaDashboard, nfaDetail, nfaList, nfaLookup) with async thunks for clean separation of concerns across 7 modules.
- **Tech Stack:** React 19, TypeScript, Vite 7, Ant Design 6, Redux Toolkit, React Quill, dayjs, SCSS, .NET API Integration, On-Prem Deployment

Project: MOC 2.0 – Management of Change Platform (Client: Nayara Energy)

- Led front-end development of an enterprise-grade Management of Change (MOC) platform for Nayara Energy's Vadinar Refinery, managing safety-critical industrial change workflows across 7 process types (Idea, DPI, PSSR, Process Change, Engineering Change, Urgent MOC, No-Cost/Soft Change).
- Designed and implemented a Component Registry pattern enabling dynamic form rendering based on stage configuration, reducing new workflow setup time by 70%.
- Built 23+ administration modules including User Role Management, Area Management, Cost Matrix, Equipment, HAZID/PHA Checklist, Email Configuration, Proxy Login, and Role Flow Management.
- Developed multi-stage approval workflows with configurable stage transitions, HAZOP/HAZID clearance gates, and role-based routing supporting 15+ approval roles.
- Integrated Module Federation (Vite Plugin Federation) for micro-frontend architecture, enabling independent deployment of the PSSR module as a remote application.
- Implemented SSO session authentication with RSA encryption; built protected route system with session persistence. Deployed on on-premises infrastructure with multi-environment build pipeline (dev/qa/vapt/migration/production).
- **Tech Stack:** React 19, TypeScript, Vite 7, Ant Design 6, Redux Toolkit, CKEditor 5, Module Federation, SCSS, SSO/RSA Auth, On-Prem Deployment

Project: LNG Deal – Global LNG Trading Platform (Client: Shell Petroleum)

- Architected and developed a global LNG contract and trading platform with 29+ view modules including Contracts, Voyages, Data Modelling, Regas Terminal Management, and Supply Project tracking for Shell's global trading desk.
- Built interactive data visualization dashboards using Highcharts, AG Grid Enterprise, and embedded PowerBI reports, enabling real-time analysis of LNG import/export flows across global markets.
- Designed visual workflow canvas using React Flow for data modelling pipelines, allowing traders to compose and configure data transformation flows via drag-and-drop.
- Implemented Atomic Design pattern (atoms → molecules → organisms) with a custom React component library (acinguiux-ds-react-framework), ensuring UI consistency across 30+ projects.
- Built OIDC-based authentication with mock OAuth2 service (RSA JWT) for development; integrated Azure Application Insights for production telemetry and performance monitoring.
- Engineered Redux + Redux Saga architecture for complex async workflows; integrated React Query for server state management alongside Redux for client state.
- Authored comprehensive Cypress component and E2E test suites; maintained SonarQube quality gates with 85%+ code coverage.
- Implemented Adobe Analytics tracking via custom React hooks, capturing user actions, page views, and routing events mapped to eVars/props for executive dashboards.
- **Tech Stack:** React 18, JavaScript, Flow, Redux, Redux Saga, React Query, AG Grid Enterprise, Highcharts, PowerBI, Ant Design 4, styled-components, Cypress, Jest, Azure App Insights, OIDC

Project: Tanso – Carbon & Energy Portfolio Dashboard (Client: Shell Petroleum)

- Designed and developed a carbon & energy trading portfolio dashboard providing real-time commodity overview across LNG, Natural Gas, and Power with global/region-level drill-down views.
- Created high-fidelity UI mockups and production screens for portfolio-level carbon tracking, login flows, and commodity-specific analytics pages.
- **Tech Stack:** React, TypeScript, Highcharts, Figma, Design System Components

Freelance Full Stack Developer

2023 – Present (Concurrent)

Independent Contractor – E-Commerce, Component Libraries & Web Applications

Project: acinguiux-ds – Enterprise React Design System (Open Source) [\[Live Docs ↗\]](#)

- Architected a multi-package React design system monorepo with 53 reusable components (Accordion, Alert, Avatar, Badge, Breadcrumb, Button, Card, Chart, DatePicker, Drawer, Dropdown, FileUploader, Grid, Modal, NavBar, Pagination, Select, Slider, Stepper, Table, Tabs, Tag, TextInput, Timeline, Toggle, Tooltip, Tree, etc.).
- Built Figma-to-code token pipeline using figma-scripts for automated design token synchronization between design and development teams.
- Published 6 NPM packages via Changesets with automated versioning: foundations (design tokens), react-base (shared utils), react-framework (53 components), react-native-framework, react-auth, and spfx-components.
- Set up monorepo infrastructure using pnpm Workspaces + Turborepo for parallel builds, Storybook 10 for component documentation, Vitest for unit testing, and Husky + ESLint + Stylelint for code quality.
- **Tech Stack:** React, TypeScript, Vite, Storybook 10, pnpm, Turborepo, Vitest, Changesets, Figma API, NPM Publishing

Project: Amitala – Preact Component Library (Open Source)

- Developed and published a lightweight Preact component library (amitala-components) with 78+ components organized by Atomic Design: 24 Atoms (button, card, form, input, table, rich-text, video), 6 Molecules (glossary, tags, tree), 48 Organisms (carousel, cart, navigation, accordion, product-admin, order-tracker).
- Built 46 Web Components (admin-dashboard, chatbot, comparison-chart, decision-tree, fuel-prices, gauge-value, product-catalogue, station-locator, universal-gallery) for framework-agnostic integration.
- Engineered JSON-driven SPA framework with page rendering engine, RTL support, i18n, and automated accessibility testing using Playwright and Siteimprove Alfa.
- Created custom esbuild toolchain for optimized production builds achieving sub-2s load times on 3G.
- **Tech Stack:** Preact 10, esbuild, Chart.js, ExcelJS, Marked, Sharp, Playwright, Web Components, NPM Publishing

Project: ShivShakti Hardware – E-Commerce Website [\[Live ↗\]](#)

- Designed and built a full-stack e-commerce website for a hardware retail store featuring product catalogues (hardware, kitchen appliances, sanitaryware, plywood, tandembox), admin panel with product management, and serverless API backend.
- Implemented multi-language support (i18n), SEO optimization (sitemap.xml, robots.txt), QR code generation, and automated accessibility testing using Playwright.
- **Tech Stack:** Preact 10 (Amitala framework), esbuild, Vercel Serverless, Vercel KV, Playwright

Project: Aurvana Interiors – Interior Design Website [\[Live ↗\]](#)

- Built a data-driven interior design showcase website with project galleries, blog, and contact form using Preact and a custom component framework (amitala-components).
- Engineered custom Node.js/esbuild SSG pipeline for static site generation, i18n support, and CDN publishing tooling; integrated Vercel serverless contact-form API with Resend email delivery.
- **Tech Stack:** Preact 10, amitala-components, esbuild SSG, Vercel Serverless, Resend API, Playwright, Chart.js

Project: FreshlyGrow – Quick Commerce Grocery Delivery App [\[Live ↗\]](#)

- Designed and built end-to-end a full-stack grocery delivery application with Preact frontend and Node.js serverless backend (14 API modules: auth, cart, orders, products, stores, users, addresses, admin, categories), deployed on Vercel with MongoDB Atlas.
- Implemented OTP-based authentication with email verification (Nodemailer), JWT sessions, bcrypt hashing, Zod validation, and real-time notifications via Pusher.
- **Tech Stack:** Preact 10, esbuild, Node.js, MongoDB Atlas, Vercel Serverless, JWT, Pusher, Nodemailer, Zod

IBM

Jul 2021 – May 2023

Advisory System Analyst

Bengaluru, Karnataka | Client: DBS Bank (Vista Equity)

- Designed Module Federation architecture for managing Micro Frontends (MFE) across DBS banking products; developed custom hooks to dynamically load remote MFE modules at runtime with shared dependency resolution across 7 country portals.
- Built cross-product common component library using styled-components with theme branding; developed DBS admin portal for real-time live chat (WebSocket) between agents and customers serving 7 countries.
- Designed Adobe Analytics (AA) tracking system using custom React hooks, mapping user actions to AA variables (props, eVars, events). Implemented PWA design compatible with native mobile applications.
- Wrote consumer API contract tests using Pact framework; developed Cypress E2E and component tests for UI automation.
- **Tech Stack:** React, TypeScript, Module Federation, styled-components, Redux, WebSocket, Adobe Analytics, Cypress, Pact, PWA

Robert Bosch Engineering and Business Solutions

Aug 2019 – Jun 2021

Senior Software Engineer

Bengaluru, Karnataka

- Developed AliveCor healthcare micro-frontend web app for provider-patient interactions: appointment booking, EKG graph analysis, FHIR data display, video/voice calls (WebRTC), and real-time chat.
- Built healthcare web apps for Walmart Health and Optimis by integrating multiple EMR/hospital management systems, handling 4,000+ clinics and 2M+ providers across the US.
- Designed Azure cloud database architecture; built Python Flask microservices for EMR integration and Azure time/data-trigger function apps for automated healthcare data pipelines. Transformed legacy EMR data to NoSQL for Patient 360 analytics.
- **Tech Stack:** React, styled-components, Python Flask, Azure Functions, CosmosDB, FHIR, WebRTC, NoSQL

EVRY India (now TietoEVRY)

Oct 2016 – Aug 2019

Software Engineer → Associate SE → Trainee

Bengaluru, Karnataka | Client: CBRE

- **Propertygram:** Built a gamification framework for property management using React Native, Firebase, and Python ML (KNN) for predicting property manager preferences. Published as IEEE research paper.
- **3-30-300 Rule:** Developed IoT + ML + VR (A-Frame) solution to optimize facility management costs; won Top Idea at Hackathon 2018.
- **CBRE Host Mobile:** Built smart building solutions app (Nike) using React Native – room booking, wayfinding, people finding, service requests; integrated CMMS, AD, and Office systems.
- **Smart Real Estate:** Developed AR/VR property tours using ARCore geolocation-based AR and A-Frame VR walkthroughs. Won Best Innovation at TECHUTSAV 2018.
- **IDA Connect:** Created NLP-powered chatbot using Android + Amazon Lex SDK with AWS Lambda for voice and chat customer support with continuous voice recognition.

- **Tech Stack:** React Native, Firebase, Python, A-Frame (VR/AR), ARCore, Android, AWS Lambda, Amazon Lex, Redux

EDUCATION

N M A M Institute of Technology, NITTE

2012 – 2016

Bachelor of Engineering – Electrical, Electronics and Communications Engineering

CERTIFICATIONS

- Scrum Fundamentals Certified (SFC)
- Digital and Data Foundations – Cloud, Technology Platforms and Cyber Security
- Digital and Data Foundations – Artificial Intelligence and Automation
- Digital and Data Foundations – Disruptive Technologies

OPEN SOURCE & NPM LIBRARIES

- **amitala-components** – Preact component library: 78+ components (Atomic Design) + 46 Web Components, JSON-driven SPA framework, RTL/i18n/a11y
- **acinguiux-ds** – React design system monorepo: 53 components, Figma token sync, Storybook docs, Turborepo, Changesets, NPM publishing

PUBLICATIONS & AWARDS

- 🌟 **Five-Star Honors "Above and Beyond"** – Recognized by Shell for quickly and efficiently delivering a full solution within a 6-month timeframe
- 🏆 **Best Innovation – TECHUTSAV 2018 (EVERY)** – Won "Smart Real Estate": AR/VR solution using ARCore and A-Frame for immersive virtual property tours
- 🏆 **Top 10 Innovation – TECHUTSAV 2019 (EVERY)** – "PropertyGram": Gamification framework for property management using React Native, Firebase, Python
- 🏆 **Top 11 – Microsoft Incubate Hackathon 2019** – 3D House Hunting using Map My India
- 💡 **Top Idea – EVERY Hackathon 2018** – "3-30-300 Rule of Facility Management": IoT + ML + VR solution for space optimization
- 🌟 **Employee of the Month (March 2018)** – EVERY India, for developing a chatbot Android application using Amazon Lex in record time
- 📄 **Publication:** Smart Real Estate – AR/VR property tour system research
- 📄 **Publication:** PropertyGram – End-to-end gamification framework for property management